

Additional file 1: Supplementary Tables S1–S5

Open, auditable processing of Visium HD spatial transcriptomics from sequencing reads to count matrices

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Table S1. Concordance with Space Ranger for every field, scale and slide. Mass ratio is STAR Suite molecule total divided by the Space Ranger total. Exact entries are the fraction of entries nonzero in either matrix that have identical counts. Correlations are Pearson r . SPATCH uses the bins in Space Ranger’s published filtered matrix at 8 μm .

Slide	Bin (μm)	Field	Mass ratio	Exact entries	Gene-total r	Bin-total r
Colorectal	2	strict	0.8815	0.8109	0.999998	0.98046
Colorectal	2	soft-expected	1.0454	0.6589	0.999999	0.98373
Colorectal	2	hard	1.0462	0.8456	0.999999	0.98430
Colorectal	2	gated-hard	0.9515	0.8197	0.999999	0.97900
Colorectal	8	strict	0.8815	0.8091	0.999998	0.99848
Colorectal	8	soft-expected	1.0454	0.8015	0.999999	0.99910
Colorectal	8	hard	1.0462	0.9045	0.999999	0.99926
Colorectal	8	gated-hard	0.9515	0.8548	0.999999	0.99877
Colorectal	16	strict	0.8815	0.7993	0.999998	0.99950
Colorectal	16	soft-expected	1.0454	0.8484	0.999999	0.99975
Colorectal	16	hard	1.0462	0.9148	0.999999	0.99980
Colorectal	16	gated-hard	0.9515	0.8551	0.999999	0.99964
Ovarian 3'	2	strict	0.8760	0.7633	0.998937	0.98186
Ovarian 3'	2	soft-expected	1.0735	0.5881	0.998913	0.98429
Ovarian 3'	2	hard	1.0764	0.8028	0.998925	0.98608
Ovarian 3'	2	gated-hard	0.9577	0.7748	0.998949	0.97943
Ovarian 3'	8	strict	0.8760	0.7427	0.998937	0.99854
Ovarian 3'	8	soft-expected	1.0735	0.7212	0.998913	0.99910
Ovarian 3'	8	hard	1.0764	0.8543	0.998925	0.99936
Ovarian 3'	8	gated-hard	0.9577	0.7975	0.998949	0.99882
Ovarian 3'	16	strict	0.8760	0.7214	0.998937	0.99943
Ovarian 3'	16	soft-expected	1.0735	0.7660	0.998913	0.99971
Ovarian 3'	16	hard	1.0764	0.8564	0.998925	0.99979
Ovarian 3'	16	gated-hard	0.9577	0.7878	0.998949	0.99961
SPATCH	8	strict	0.8054	—	0.999995	0.99663
SPATCH	8	soft-expected	1.0229	—	0.999997	0.99841
SPATCH	8	hard	1.0232	—	0.999997	0.99877
SPATCH	8	gated-hard	0.8876	—	0.999996	0.99756

Table S2. SPATCH slide processed with the 2024-A reference. The published Space Ranger matrix uses the 2020-A probe set (18,085 features); the union of the gene sets has 18,536 features. Unmatched mass counts molecules on genes present in only one output. Gene-total Pearson r uses shared genes; bin-total Pearson r uses the 8 μm bins in Space Ranger’s published matrix.

Field	STAR features	Shared	STAR mass, unmatched	SR mass, unmatched	Mass ratio	Gene-total r	Bin-total r
strict	18,417	17,966	2,942,415	225,090	0.8103	0.988145	0.99548
soft-expected	18,417	17,966	3,733,968	225,090	1.0291	0.988130	0.99723
hard	18,417	17,966	3,735,012	225,090	1.0294	0.988131	0.99758
gated-hard	18,417	17,966	3,241,867	225,090	0.8930	0.988113	0.99639

Table S3. CODEX comparison on the SPATCH slide. Mean ROC AUC across 14 matched RNA–protein markers. For each marker, protein-positive squares are those at or above the 80th percentile of CODEX intensity, on grids from 100 to 500 μm .

Field	100 μm	200 μm	300 μm	400 μm	500 μm
Space Ranger (2020-A)	0.5980	0.6241	0.6359	0.6269	0.6640
STAR strict (2020-A)	0.5952	0.6208	0.6354	0.6261	0.6643
STAR soft-expected (2020-A)	0.5978	0.6232	0.6351	0.6265	0.6640
STAR hard (2020-A)	0.5976	0.6233	0.6351	0.6264	0.6641
STAR gated-hard (2020-A)	0.5967	0.6228	0.6360	0.6266	0.6660
STAR strict (2024-A)	0.5952	0.6208	0.6354	0.6261	0.6642
STAR soft-expected (2024-A)	0.5978	0.6231	0.6351	0.6265	0.6640
STAR hard (2024-A)	0.5976	0.6232	0.6351	0.6264	0.6641
STAR gated-hard (2024-A)	0.5967	0.6228	0.6360	0.6266	0.6660

Table S3 (continued). Marker mass shared and added (2020-A, hard). RNA counts for the matched markers in the scored squares. Shared counts are the smaller count from the two pipelines in each square; each pipeline’s remainder is its residual. These are differences in counts, not matched molecules.

Grid (μm)	Space Ranger	STAR	Shared	STAR only	Space Ranger only
100	794,704	811,385	792,536	18,849	2,168
200	795,980	812,891	795,116	17,775	864
300	795,953	812,969	795,473	17,496	480
400	796,002	813,310	795,717	17,593	285
500	796,000	813,209	795,817	17,392	183

Table S4. Registration sensitivity on the colorectal slide. Each entry gives the score with Space Ranger’s microscope registration, followed by the score with our registration, using the same image masks reprojected through each. Reprojection changes 3.07% of grid squares’ cell-mask state and 15.39% of their nucleus-mask state (median displacement 2.70 μm). The largest change in any STAR minus Space Ranger score difference is 1.97×10^{-3} .

Field	In-cell fraction	Cell ROC AUC	Nucleus ROC AUC
Space Ranger	0.9494 / 0.9483	0.8522 / 0.8511	0.6980 / 0.7048
STAR strict	0.9498 / 0.9487	0.8485 / 0.8474	0.6963 / 0.7035
STAR hard	0.9491 / 0.9481	0.8549 / 0.8537	0.6996 / 0.7064
hard, STAR only	0.9429 / 0.9417	0.6751 / 0.6743	0.6133 / 0.6172
hard, Space Ranger only	0.9428 / 0.9416	0.5983 / 0.5978	0.5641 / 0.5655

Table S5. Ovarian 3’ slide under the two multimapper settings. Molecule totals over the full capture grid and percentage differences from Space Ranger 4.1.0 with introns included (187,552,358 molecules).

Field	Compatibility	vs SR (%)	Annotated	vs SR (%)
strict	164,293,957	-12.40	155,685,146	-16.99
soft-expected	201,338,571	7.35	190,819,513	1.74
hard	201,888,275	7.64	191,328,545	2.01
gated-hard	179,610,526	-4.23	170,196,357	-9.25

Table S5 (continued). Xenium endpoints under the two settings. Pearson correlation with Xenium of cancer-rich versus cancer-poor effects across the cancer panel, excluding the label genes and MECOM, and median spatial Pearson correlation for the most abundant Xenium panel genes. The latter gene set is fixed by Xenium abundance before comparing Visium fields; its size is given in the column headings.

Field	Effect r , compat.	Effect r , annot.	Top-20 r , compat.	Top-20 r , annot.
Space Ranger	0.857	0.857	0.238	0.238
strict	0.863	0.863	0.237	0.236
hard	0.866	0.867	0.257	0.256